



KLAUS BARKHAUSEN

Engineering Leader | Enterprise Architect | AI-Augmented Delivery

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Hands-on engineering leader with 20+ years building and shipping enterprise and subscription SaaS on Azure and .NET. Proven player-coach who sets architecture, writes production code, and raises team standards across distributed remote, onshore, and offshore engineers through code review, TDD, and modern CI/CD practice. Deep, demonstrated AI-augmented delivery on two distinct axes: AI-assisted coding (Claude Code, Cursor, and agentic workflows—including authoring a practitioner book on Claude Code), and embedding AI intelligence directly into both legacy and greenfield codebases so systems act more autonomously, make decisions at human-level quality, and accelerate business outcomes. Track record across microservices, distributed and event-driven architectures (Azure Service Bus, Outbox, CQRS/MediatR), and the re-engineering of legacy .NET platforms. Bilingual (English/Spanish).

CORE COMPETENCIES

Engineering Leadership

- Player-coach delivery — ship code & lead team
- Enterprise architecture, re-platforms & legacy re-engineering
- Code review, TDD, CI/CD & on-call hygiene
- Distributed remote, onshore & offshore team leadership
- Roadmap shaping with Product leadership
- Engineering spend, vendors & tooling strategy

Cloud, DevOps & CI/CD

- CI/CD: GitHub Actions, Azure DevOps / Pipelines, YAML
- Containers & orchestration: Docker, Kubernetes (AKS)
- Infrastructure as Code: Bicep, ARM, Terraform
- Observability: OpenTelemetry, Application Insights
- Agentic CI/CD: issue-to-PR pipelines, autonomous dev loops

Technical Depth

- Cloud: Azure, AWS, Google
- .NET: ASP.NET Core, EF Core, CQRS/MediatR, xUnit
- AI-assisted dev: Claude Code, Cursor, Copilot, MCP
- Microservices, distributed systems, Outbox pattern, CQRS
- APIs: REST, Web API, OpenAPI
- Data stores: SQL Server, PostgreSQL, Cosmos DB, Azure Table Storage, Redis, Neo4j
- Vector & search: Azure AI Search, Pinecone, Elasticsearch
- AI: Semantic Kernel, AWS Bedrock, Azure AI Document Intelligence, Azure AI Foundry

Programming Languages

- C#, JavaScript, TypeScript, Python
- SQL / T-SQL, KQL (Kusto)
- PowerShell, Bash (shell scripting)
- HTML5, CSS3, JSON, YAML
- Frontend: React, Angular, Vue.js, Node.js

ENGINEERING FOCUS AREAS

AI-Augmented Engineering

AI-assisted code generation • Embedding LLMs & agents in legacy and new systems • Autonomous & human-in-the-loop decisioning • Internal AI tooling strategy

App Dev & Modernization

ASP.NET Core re-platforms • Legacy .NET modernization • Event-driven architecture • Microservices & APIs

Test Automation & QE

Test automation frameworks • CI/CD quality gates • Performance testing • Shift-left QA

CERTIFICATIONS & EDUCATION

AWS AI Practitioner | Microsoft Azure AI Fundamentals | Google AI Essentials

B.S. Business Administration — Nova Southeastern University, Miami, FL

Languages: English and Spanish — fluent in both

PROFESSIONAL EXPERIENCE

Principal Engineer & AI Solutions Architect

Feb. 2026 – Present

5 STARS TECHNOLOGY, Miami, FL

Founder of a boutique AI engineering practice serving mid-market SaaS, hospitality, healthcare, and logistics clients across South Florida. Operate as a hands-on principal engineer—shipping production .NET / Azure systems, leading small onshore and offshore teams, and integrating AI-assisted development workflows that materially increase team output. Own enterprise architecture, vendor and cloud-spend decisions, and AI tooling strategy across engagements.

Lead architect and player-coach on Avolta, a perfume-tester shipment-tracking platform for International Operations Services (UY) S.A. Designed and shipped the architecture from scratch on ASP.NET Core, Vue.js 3, Azure Service Bus with the Outbox pattern, EF Core, CQRS/MediatR, SQL Server, and xUnit—an event-driven microservices design with strong test coverage and CI/CD on Azure DevOps. Produce production code directly while coordinating offshore contributors via AI-assisted review loops and shared CLAUDE.md specifications.

Beyond AI-assisted coding, design and ship intelligent features inside client systems—embedding LLMs, classifiers, and decision agents into both legacy and greenfield codebases so software acts more autonomously, makes decisions at human-level quality, and frees teams to focus on higher-value work. Recent work includes Azure AI Document Intelligence pipelines (hybrid classification plus prebuilt extraction with confidence scoring and human-in-the-loop routing), vendor matching using fuzzy and semantic similarity, and multi-agent workflows wired into Azure Service Bus event flows. Translate these capabilities into measurable business outcomes—lower cycle time, fewer escalations, and reliable autonomous decisioning on tasks previously handled by people.

Authoring a practitioner book on Claude Code (Anthropic's coding agent), structured around a running project that scales from beginner to advanced. Run deep AI-augmented development workflows in real engagements—autonomous iterative dev loops, GitHub Actions agentic issue-to-PR pipelines, modular CLAUDE.md / .claude/rules/ architecture, and MCP server integration—with measured impact on shipping velocity. Concrete, hands-on AI tooling strategy, not theory.

AI Solutions Lead / .NET Architect / Cloud & TDD Evangelist

Apr. 2014 – Feb. 2026

ANTHOLOGY, Boca Raton, FL

Hands-on .NET architect and player-coach on enterprise SaaS platforms serving higher-education institutions globally—a subscription, multi-tenant product portfolio with deep customer-journey workflows from onboarding through expansion. Partnered with product leadership to set technical roadmap, sequence platform improvements, and ship parallel with new product work.

Integrated AI/LLM patterns (Semantic Kernel, AWS Bedrock, RAG) into production product features and championed AI-assisted coding adoption across the engineering org. Established team-wide standards for clean architecture, TDD, CI/CD quality gates, and observability (OpenTelemetry, Application Insights), raising shipping cadence and reducing on-call burden.

Wrote production code on the core platform while leading architecture for major features—sequencing the work so new product roadmap commitments shipped in parallel with platform improvements. Served as the technical bridge between product strategy and day-to-day engineering execution.

Sr. Solution Architect & Team Leader

Apr. 2013 – Apr. 2014

AVISENA, Miami, FL

Led architecture and team for the re-engineering of a legacy multi-tenant SaaS healthcare platform onto modern .NET. Designed the target architecture using clean boundaries, domain-driven design, and modern patterns; drove implementation hands-on with the engineering team while establishing testing and review standards that reduced defects and accelerated delivery.

Solutions Architect & Technology Consultant

Mar. 2006 – Apr. 2013

INDEPENDENT PRACTICE, Miami, FL

Operated as principal engineer and architect for enterprise clients across retail, financial services, and luxury goods. Owned solution design, implementation, and delivery on multi-stakeholder programs—hands-on across .NET, integration, and data layers—while leading distributed remote and offshore contributor pools and managing scope, timelines, and stakeholder communications.

The following are representative engagements with major global enterprises, illustrating the scope and level of client relationships managed throughout this period:

Cartier (Latin America) — Global luxury brand, HQ Switzerland, 4K+ employees, ~\$7B annual revenue: Designed and delivered a regional reporting and operations platform that reduced sales-data consolidation from 45–60 days to 1–3 days, enabling near real-time decision-making. Owned architecture and implementation end-to-end; solution was recognized by global operations and expanded to additional regions.

McAfee (Latin America) — Global cybersecurity leader, presence in 180+ countries, 7,000+ employees: Designed and shipped a telecom-channel integration platform enabling a new revenue-generating bundled-product model. Owned integration architecture and partner-system contracts end-to-end, producing a 40% increase in customer satisfaction and 32% increase in bundled product sales.

Richemont IT USA — Global luxury goods conglomerate, HQ Switzerland, 35K+ employees, ~\$20B annual revenue: Built an enterprise integration linking social-media platforms to core business systems, enabling marketing to attribute conversions to specific campaigns and substantially improve marketing investment decisions.

Additional engagements for Aflac, Ritz-Carlton, and others across retail, healthcare, and financial services.

Regional I.T. Manager / Sr. Programmer

May 2004 – Mar. 2006

RICHEMONT LATIN AMERICA & CARIBBEAN LLC, Miami, FL

Led end-to-end IT operations for Latin America & Caribbean, owning strategy, budgets, and delivery across infrastructure and software. Directed a cross-functional team as a player-coach, instituting lean practices that increased productivity ~40% while operating with ~30% less budget. Managed vendor relationships and stakeholder communications across the region.

Sr. Programmer / Project Manager

Sept. 2000 – May 2004

DIGITALERA GROUP LLC, Miami, FL

Conceived, architected, and delivered a B2B solution for Cartier Latin America to automate retail operations across the region. Gathered requirements, wrote specifications, designed the UI, and managed an offshore development team to lower costs.

Sr. Programmer / Project Manager

Jan. 2000 – Sept. 2000

TRAPEZOID INC., Miami, FL

Lead architect for all software projects across onshore and offshore teams. Architected and delivered solutions for start-up and mid-to-large organizations.

Sr. Programmer / Oracle DBA

Sept. 1998 – Dec. 1999

CHS ELECTRONICS INC., Miami, FL

Senior member of the team responsible for creating a sophisticated e-commerce solution serving 4,000+ customers. Developed a self-service web portal that cut customer service operations costs by over 60%.

PUBLICATIONS

Total Fitness with Artificial Intelligence — Practical Fitness and Cutting-Edge AI

A practitioner guide to applying AI to the four pillars of health and fitness—mindset, nutrition, exercise, and recovery—with ChatGPT-powered personalized coaching workflows. Published October 2023, paperback and Kindle, 275 pages. Companion website: totalfitness.ai